MAT 322 Spring 2025

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Contents

1 Week 1: 1/27 and 1/29

1.1 Topology of Euclidean space (Munkres §3)

Definition 1.1 (Euclidean n-space) — **Euclidean n-space**, denoted $\mathbb{R}^n$, is the set of all n-tuples of real numbers.

$\mathbb{R}^n$ is a vector space over $\mathbb{R}$: it is equipped with two binary operations, addition and scalar multiplication. We then care about functions that behave nicely with these two operations:

Definition 1.2 — A function $f\colon \mathbb{R}^n \to \mathbb{R}^m$ is **linear** if it respects the addition and scalar multiplication operations — that is, $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}^n$ and $f(cx) = cf(x)$ for all $c \in \mathbb{R}$ and $x \in \mathbb{R}^n$.

More tangibly, we can say that if $e_1, \ldots, e_n$ is a basis for $\mathbb{R}^n$, then $f(e_1), \ldots, f(e_n)$ determine the values of the function on every other point of $\mathbb{R}^n$, since each point of $\mathbb{R}^n$ is a linear combination of the e_i. We will generally reserve $e_1, \ldots, e_n$ for the **standard basis** of $\mathbb{R}^n$, i.e. the unit vectors along the positive direction of each axis (so $e_3 = (0, 0, 1, 0, \ldots, 0)$). If we also specify a basis for the codomain $\mathbb{R}^m$ (which by abuse of notation we will just reuse $e_1, \ldots, e_m$; the ambient space should be clear by context), then we can say that

$$\begin{aligned} f(e_1) &= a_{11}e_1 + a_{21}e_2 + \cdots + a_{m1}e_m \\ &\vdots \\ f(e_n) &= a_{1n}e_1 + a_{2n}e_2 + \cdots + a_{mn}e_m, \end{aligned}$$

and we represent f with the matrix

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}.$$

However, $\mathbb{R}^n$ isn't just any old vector space; we can measure length and angles, too (unlike, say, $\mathbb{R}[x]$). Abstractly, the linear algebraic mechanism for this is the inner product $\langle \bullet, \bullet \rangle\colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ defined by

$$\langle x, y \rangle = \sum_{i=1}^{n} x_i y_i.$$

Then, $\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{x_1^2 + \cdots + x_n^2}$ is the **Euclidean norm** on $\mathbb{R}^n$. The norm induces a metric on $\mathbb{R}^n$ by $d(x, y) = \|x - y\|$. Specifically, one can verify the following proposition:

Proposition 1.3 (d is a metric). For all $x, y, z \in \mathbb{R}^n$, the function $d\colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ satisfies

(a) Symmetry: $d(x, y) = d(y, x)$.

(b) Positive definiteness: $d(x, y) \geq 0$ with equality iff $x = y$.

(c) Triangle inequality: $d(x, z) \leq d(x, y) + d(y, z)$.

Proof. We only prove the triangle inequality. We want to show that

$$\|x - z\| \leq \|x - y\| + \|y - z\|.$$

After substituting $a = x - y$ and $b = y - z$, this becomes $\|a + b\| \leq \|a\| + \|b\|$. Squaring gives the equivalent

$$\langle a + b, a + b \rangle \leq \langle a, a \rangle + \langle b, b \rangle + 2\|a\| \cdot \|b\|.$$

Expanding the LHS via bilinearity reduces to

$$\langle a, b \rangle \leq \|a\| \cdot \|b\|,$$

the Cauchy-Schwarz inequality, which we take for granted (but you can square, expand, and spam AM-GM). □

Remark 1.4. Actually, the variation of the triangle inequality with a and b that I proved is the triangle inequality that norms are required to satisfy, so this would've been free if we verified that $\| \bullet \|$ is actually a norm on $\mathbb{R}^n$.

This metric induced by the norm gives us the natural metric topology on $\mathbb{R}^n$.

Definition 1.5 — The **open ball** centered at x with radius r is the set

$$B_r(x) = \{y \in \mathbb{R}^n : d(x, y) < r\}.$$

Then, we say that $U \subseteq \mathbb{R}^n$ is **open** in $\mathbb{R}^n$ if, for each $x \in U$, some open ball centered at x is contained in U. A subset $C \subseteq \mathbb{R}^n$ is **closed** (in $\mathbb{R}^n$) if its complement $\mathbb{R}^n \setminus C$ is open (in $\mathbb{R}^n$).

Example 1.6

Open intervals (a, b) are open in $\mathbb{R}$, and indeed a product of n intervals $(a_1, b_1) \times \cdots \times (a_n, b_n)$ is an open subset of $\mathbb{R}^n$.

Similarly, products of closed intervals $[a_1, b_1] \times \cdots \times [a_n, b_n] \subseteq \mathbb{R}^n$ are closed, as is the **closed ball** $\overline{B_r(x)} := \{y \in \mathbb{R}^n : d(x, y) \leq r\}$.

On the other hand, $(a, b]$ is neither open nor closed in $\mathbb{R}$. **Open and closed are not opposites in topology**.

Definition 1.7 (Continuity) — Let $U \subseteq \mathbb{R}^n$ be open and $f : U \to \mathbb{R}^m$ a function. Then, f is **continuous** at a point $x \in U$ if given any $\varepsilon > 0$, there exists $\delta > 0$ such that $d(x, y) < \delta$ implies $d(f(x), f(y)) < \varepsilon$. If f is continuous at every point in its domain then we can use "continuous" to describe f itself, and if f is continuous at every point in $V \subseteq U$ then we say that f is continuous on V.

We can also phrase this definition more globally:

Proposition 1.8. Let $U \subseteq \mathbb{R}^n$. Then $f : U \to \mathbb{R}^m$ is continuous if and only if, for each open $V \subseteq \mathbb{R}^m$, the pre-image $f^{-1}(V) \subseteq \mathbb{R}^n$ is also open.

Proposition 1.9. A linear function $f : \mathbb{R}^n \to \mathbb{R}^m$ is continuous.

Proof Sketch. Just write $f(x) = Ax$ for some matrix A and then expand $\|f(x) - f(y)\|^2$ in terms of the entries of A to get a bound. □

1.2 Differentiation

Differentiation is about approximating continuous functions by linear functions.

Definition 1.10 — Let $U \subseteq \mathbb{R}^n$ be open and let $f\colon U \to \mathbb{R}^m$ be a function. We say that f is **differentiable** at $x \in U$ if there exists a linear function $L\colon \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{h \to 0} \frac{f(x+h) - f(x) - L(h)}{\|h\|} = 0,$$

and we plainly say that f is differentiable if it is differentiable at every point in U. The **derivative** of f at x is this linear function L, and we denote this by $(\mathrm{d}f)_x$.

Remark 1.11. We can rephrase this as saying for any $\varepsilon > 0$, there exists a $\delta > 0$ such that $0 < \|h\| < \delta$ implies

$$\|f(x+h) - f(x) - L(h)\| < \varepsilon \|h\|,$$

unfurling the limit definition.

Compare this to the standard definition of differentiability with a function $f\colon \mathbb{R} \to \mathbb{R}$; there are a few discrepancies. The one I will point out is that the "derivative" in the sense that we use it in single variable calculus is a number rather than a function because linear functions $\mathbb{R} \to \mathbb{R}$ can be identified with their slope.

Exercise 1.12. Let $f\colon \mathbb{R}^n \to \mathbb{R}^m$ be defined by $f(x) = Ax + b$ where A is an $m \times n$ matrix and $b \in \mathbb{R}^m$. What is the derivative of f? (for the moment, take for granted that the derivative is unique)

Proposition 1.13. If $U \subseteq \mathbb{R}^n$ is open and $f\colon U \to \mathbb{R}^m$ is differentiable at x, then f is also continuous at x.

Exercise 1.14. Prove this (it is the exact same proof as the $\mathbb{R} \to \mathbb{R}$ case).

Definition 1.15 — The **directional derivative** of f at $x \in U$ along $v \in \mathbb{R}^n$ is the vector

$$\frac{\partial f}{\partial v}(x) := \lim_{t \to 0} \frac{f(x+tv) - f(x)}{t}.$$

If $e_1, \ldots, e_n \in \mathbb{R}^n$ is the standard basis, we instead write

$$\frac{\partial f}{\partial x_i}(x) := \frac{\partial f}{\partial e_i}(x).$$

Proposition 1.16. If f is differentiable at $x \in U$, then $\frac{\partial f}{\partial v}$ exists for all $v \in \mathbb{R}^n$. Moreover,

$$(\mathrm{d}f)_x(v) = \frac{\partial f}{\partial v}(x).$$

Remark 1.17. This implies that the derivative of f is unique, as the wording of the definition of derivative suggests.

left as an exercise for us

Proof. $\square$

Warning 1.18 — The proposition essentially says that a differentiable function at x has all its partial derivatives at x. One might conjecture the converse statement that having all partial derivatives implies f is differentiable, but this is false, as the example of

$$f(x,y) = \begin{cases} \frac{x^3}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

shows: all directional derivatives exist everywhere but f isn't differentiable at $(0,0)$.

The function $(\mathrm{d}f)_x$, being linear, is a matrix called the **Jacobian matrix**. The preceding proposition lets us say that if

$$f = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}, \quad f_j \colon \mathbb{R}^n \to \mathbb{R}$$

then

$$(\mathrm{d}f)_x = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{x_n} \end{pmatrix}.$$

Definition 1.19 — A function $f \colon U \to \mathbb{R}^m$ is **continuously differentiable** (or **class C^1**) on U if $\partial f / \partial x_i$ exists at all $u \in U$ and $i = 1, 2, \ldots, n$ and

$$\frac{\partial f}{\partial x_i} \colon U \to \mathbb{R}^m$$

is continuous.

The nice thing about this is the following fact:

Theorem 1.20. If $f \colon U \to \mathbb{R}^m$ is C^1 then it is differentiable.

pf

Proof. □

2 Week 2: 2/3 and 2/5

2.1 Chain rule

Example 2.1 (1-D chain rule)

Suppose we have functions $f \colon (a,b) \to \mathbb{R}$ and $g \colon (c,d) \to \mathbb{R}$ such that

- f is differentiable at $u \in (a,b)$.
- g is differentiable at $v = f(u) \in (c,d)$.

Then, the composition $(g \circ f)(x) = g(f(x))$ is differentiable at $x = 0$ and $(g \circ f)'(u) = g'(f(u))f'(u) = g'(v)f'(u)$ (sweeping domain issues under the rug).

The generalization of this to several variables is as follows:

Theorem 2.2 (*n*-D chain rule)**.** Let $f\colon U \to \mathbb{R}^m$ for an open $U \subseteq \mathbb{R}^n$ and let $g\colon V \to \mathbb{R}^\ell$ for an open $V \subseteq \mathbb{R}^m$. Suppose that

- f is differentiable at $u \in U$.
- g is differentiable at $v = f(u) \in V$.

Then, $g \circ f$ is differentiable at u and

$$(\mathrm{d}(g \circ f))_u = (\mathrm{d}g)_v \circ (\mathrm{d}f)_u \colon \mathbb{R}^n \to \mathbb{R}^\ell.$$

Remark 2.3. This makes sense in 1 dimension because composition of linear maps $\mathbb{R} \to \mathbb{R}$ looks like multiplying their coefficients.

You can also write this out by multiplying the Jacobians but i don't hate myself ♡

Proof. Following the definition of derivative,

(1) $f(u+h) - f(u) = (\mathrm{d}f)_u(h) + \|h\|F(h)$, and $\lim_{h\to 0} F(h) = 0$.

(2) $g(v+k) - g(v) = (\mathrm{d}g)_u(h) + \|k\|G(k)$, and $\lim_{h\to 0} G(h) = 0$.

Then, letting $\Delta(h) = f(u+h) - f(u)$,

$$\begin{aligned}(g \circ f)(u+h) - (g \circ f)(u) &= g(f(u+h)) - g(f(u)) \\ &= g(v+k) - g(v) \\ &= (\mathrm{d}g)_v(\Delta) + \|k\|G(k) \\ &= (\mathrm{d}g)_v \circ (\mathrm{d}g)_u + \|h\|(\mathrm{d}g)_v F(h) + \|\Delta\|G(\Delta).\end{aligned}$$

finish

□

Theorem 2.4 (Inverse function theorem)**.** If $f\colon U \to \mathbb{R}^n$ for open $u \subseteq \mathbb{R}^n$ is a C^1 function such that $(\mathrm{d}f)_u \colon \mathbb{R}^n \to \mathbb{R}^n$ is invertible for some $u \in U$, there exists an open $U' \subseteq U$ containing u such that $V' = f(U')$ is open and there is a C^1 function $g\colon V' \to U'$ such that

$$g \circ (f|_{U'}) = \mathrm{Id}_{U'}, \quad f \circ g = \mathrm{Id}_{V'}.$$

Proof. Let f be as in the statement.

Lemma (Injectivity)**.** There exist constants $\delta > 0$, $\alpha > 0$ such that if $x_0, x_1 \in B_\delta(u)$, then $\|f(x_1) - f(x_0)\| \geq \alpha \cdot \|x_1 - x_0\|$.

todo

Proof. □

In particular, f is injective on $B_\delta(u)$, since there are no two distinct points x_0, x_1 such that $\|f(x_1) - f(x_0)\| = 0 < \alpha \cdot \|x_1 - x_0\|$.

Lemma (Open to open)**.** Let $B_\delta(u) \subseteq U$ be as in the previous lemma. Then $f(B_\delta(u)) \subseteq \mathbb{R}^m$ is open.

Proof. Let $x \in B_\delta(u)$ and $y = f(x) \in f(B_\delta(u))$. Then, we are looking for an $\varepsilon > 0$ such that $B_\varepsilon(y) \subseteq f(B_\delta(u))$.

Let $r < \delta$ be such that $y \in f(B_r(u))$. Since f is injective on $B_\delta(u)$, y is not contained in $f(\partial B_r(u))$, the image of the boundary of $B_r(u)$. The magic trick now is that $\partial B_r(u)$ is compact, so its image is compact hence closed (because $\mathbb{R}^n$ is Hausdorff); this means we can find a neighborhood of y, say $B_\varepsilon(y)$, that doesn't intersect $f(\partial B_r(u))$.

It remains to prove that $B_\varepsilon(y)$ is contained in $f(B_r(u))$. That is, we want to show that any given $y_0 \in B_\varepsilon(y)$ is contained in $f(B_r(u))$. Consider a function $\varphi\colon B_\delta(u) \to \mathbb{R}$ given by $\varphi(z) = \|f(z) - y_0\|^2$. We know that φ is continuous, so it attains a minimum on the compact set $\overline{B_r(u)}$ at some point $x_0 \in \overline{B_r(u)}$. We know a priori that

$$\varphi(x) = \|f(x) - y_0\|^2 = \|y - y_0\|^2 < \varepsilon^2,$$

so $x \notin \partial B_r(u)$ because $f(\partial B_r(u))$ is disjoint from $B_\varepsilon(y)$. Thus, $x_0 \in B_r(u)$. Then we can apparently use the first lemma from lesson 4 to conclude that $(\mathrm{d}\varphi)_{x_0} = 0$, but then we also know that

$$0 = (\mathrm{d}\varphi)_{x_0}(h) = 2\langle f(x_0) - y_0, (\mathrm{d}f)_{x_0}(h)\rangle.$$

We must then have $f(x_0) - y_0$ is the zero vector, and we can use injectivity to conclude $y_0 = f(x_0)$, which is contained inside $f(B_r(u))$.

> **Remark.** Is it true that $f(\partial B_r(u)) = \partial f(B_r(u))$? If so, a proof involving the fact that continuous functions preserve connectedness would go through.

□

Now let $U' = B_\delta(u)$; thus f is a bijection $U' \to V' := f(B_\delta(U)) = f(U')$. Thus there is a unique $g\colon V' \to U'$ that is the inverse of f on U'. We know that $\det(\mathrm{d}f)_x \neq 0$ holds on a neighborhood of x because the partials of f are continuous (Proof: we have a polynomial condition in the partials, so the locus of points for which the polynomial is zero will be a closed set, thus having open complement), so shrink δ even further so that we now live in a neighborhood where the determinant is always nonzero (we'll still call the shrunk number δ).

> **Claim —** g is continuous.

Proof. We can use the "open to open" lemma to argue that the image of any open subset of $B_\delta(u)$ under f is open in $\mathbb{R}^m$, so f is an open map on U' hence its inverse g is continuous (this is why we needed to make sure that the derivative is invertible on all of U'). □

Now, we will show that $g\colon V' \to U'$ is differentiable. Let $y \in V'$ and $g(y) = x$. Also let $A = (\mathrm{d}f)_x$ and $\Delta(k) = g(y+k) - g(y)$ for $k \neq 0$ (so that $\Delta(k)$ is never zero). Our goal is to show that $(\mathrm{d}g)_x = A^{-1}$. We can write

$$k = f(x + \Delta(k)) - f(x),$$

so that

$$\begin{aligned}\frac{g(y+k) - g(y) - A^{-1}k}{\|k\|} &= \frac{\Delta(k) - A^{-1}k}{\|k\|} \\ &= -A^{-1} \cdot \frac{k - A\Delta(k)}{\|k\|} \\ &= -A^{-1} \cdot \frac{f(x+\Delta(k)) - f(x) - A\Delta(k)}{\|\Delta(k)\|} \cdot \frac{\|\Delta(k)\|}{\|k\|}.\end{aligned}$$

We know that g is continuous, so as $k \to 0$, $\Delta(k) \to 0$, hence the whole thing in the middle goes to 0. To show that the entire expression goes to zero (thereby proving that $(\mathrm{d}g)_x = A^{-1}$), we only need to show that $\frac{\|\Delta(k)\|}{\|k\|}$ is bounded. Luckily, this is exactly what the injectivity lemma above proves, so g is differentiable.

Finally, before we can go home, we have to prove that g is in fact *continuously* differentiable. This follows from the entries of the Jacobian matrix of g at $f(x)$ being "algebraic functions" of the entries of $(\mathrm{d}f)_x^{-1}$, which are themselves continuous functions of x (or matrix inversion is continuous on the space of invertible matrices). And the crowd goes wild. $\square$

3 Week 3: 2/10 and 2/12

The first lesson was completing the proof of the inverse function theorem.

We now talk about a consequence of the inverse function theorem.

3.1 Implicit function theorem

Let's look at a small example. Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be a C^1 function.

Question. When does $f(x,y) = 0$ determine y implicitly as a function of x?

Example 3.1

Let $f(x,y) := x^2+y^2-1 = 0$. This locus of points is what's commonly known as a circle. Then, if you look at a point in this locus and consider some open neighborhood, you can usually find y as a function of x; for example if you have a small enough neighborhood of $(3/5, 4/5)$ then $y = \sqrt{1-x^2}$, or if your point is $(7/25, -24/25)$ then $y = -\sqrt{1-x^2}$ on a small neighborhood.

On the other hand, you run into issues at $(1,0)$ because you have points with both positive and negative y (and same with $(-1,0)$). Concretely, the issue here is that $\partial f/\partial y(\pm 1, 0) = 0$. We can, though, write x as a function of y in this case.

As usual we care very much about local behavior and pretty much not at all about global behavior.

In general, let $f\colon U \to \mathbb{R}^m$ be C^1 where $U \subseteq \mathbb{R}^n \times \mathbb{R}^n = \mathbb{R}^{n+m}$ is open. Then, write

$$\mathrm{d}f = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} & \frac{\partial f_1}{\partial y_1} & \cdots & \frac{\partial f_1}{\partial y_m} \\ \vdots & & & & & \\ \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} & \frac{\partial f_1}{\partial y_1} & \cdots & \frac{\partial f_1}{\partial y_m} \end{pmatrix}$$

how do you make line down the middle of matrix?

and write $\partial f/\partial x$ for the left half of the matrix and $\partial f/\partial y$ for the right half. (so instead of writing x_{n+k} we write y_k for the latter m coordinates for convenience)

Theorem 3.2. If $f\colon U \subseteq \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ be a C^1 function. Suppose $\partial f/\partial y$ is invertible at $(x_0, y_0) \in U$ and $f(x_0, y_0) = 0$. Then, there exist some open A and B containing x_0 and y_0 such that there is a unique C^1 function $g\colon A \to B$ satisfying

$$f(x, g(x)) = 0, \quad g(x_0) = y_0.$$

The idea is that the derivative being invertible corresponds to the idea of never having a "vertical tangent line" in a neighborhood of (x_0, y_0). In particular we can project points in $\{f(x,y) = 0\}$ to the "x-axis" ($\mathbb{R}^n$) [in an injective fashion wrt the graph?], which gives you a C^1 parametrization of the locus near (x_0, y_0).

Now we shall write down an actual proof.

Proof. Define $F\colon U\colon \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n \times \mathbb{R}^m$ by $F(x,y) = (x, f(x,y))$. The derivative at (x,y) looks like

$$(\mathrm{d}F)_{(x,y)} = \begin{pmatrix} I_n & 0 \\ \frac{\partial f}{\partial x}(x,y) & \frac{\partial f}{\partial y}(x,y) \end{pmatrix}.$$

Then this is invertible because $\frac{\partial f}{\partial y}$ is by assumption invertible (either due to determinant calculations or due to row reduction).

By the inverse function theorem, we can find an open $U' \subseteq U$ containing (x_0, y_0) such that $V' = F(U')$ is open and there is a C^1 inverse function $G\colon V' \to U'$ of F within these small open sets. We can assume that $U' = U_1' \times B$ is a product of open sets such that $x_0 \in U_1'$ and $y_0 \in B$.

Write $G(x,y) = (P(x,y), Q(x,y)) \in \mathbb{R}^n \times \mathbb{R}^m$. Then

$$G \circ (F|_{U'}) = \mathrm{Id}_{U'}$$

means that $P(x, f(x,y)) = x$ and $Q(x, f(x,y)) = y$. Similarly

$$(F|_{U'}) \circ G = \mathrm{Id}_{V'}$$

entails $P(x,y) = x$ and $f(P(x,y), Q(x,y)) = y$.

Let $A \times V_2' \subseteq V'$ be a product of open sets where $x_0 \in A$ and $0 \in V_2'$. Then consider $g\colon A \to B$ given by $g(x) = Q(x,0)$. We can see that

$$f(x, g(x)) = f(P(x,0), Q(x,0)) = 0$$

and, from $Q(x, f(x,y)) = y$, we can see that if $f(x,g(x)) = 0$ then $Q(x,0) = Q(x, f(x, g(x))) = g(x)$, i.e. this $g := Q(x,0)$ is really the only function we can procure (and checking g is C^1 is easy). $\square$

Exercise 3.3. Compute the derivative of g at x_0.

4 Week 4: 2/17 and 2/19

2/17 notes

4.1 Differentiate, and then do it again

Theorem 4.1 (Taylor's theorem)**.** Let $f\colon U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be a C^k function and let $x \in U$. Then, the Taylor polynomial of f centered at x is the polynomial

$$f(x+h) = \sum_{|\alpha| \le k} \frac{1}{\alpha!} \frac{\partial^\alpha f}{\partial x^\alpha}(x) \cdot h^\alpha + o(h^k).$$

Here α is a tuple $(\alpha_1, \ldots, \alpha_n)$ of nonnegative integers, and we define

$$\alpha! = \alpha_1! \cdots \alpha_n!, \quad \frac{\partial^\alpha}{\partial x^\alpha} = \left(\frac{\partial}{\partial x_1}\right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_n}\right)^{\alpha_n}, \quad h^\alpha = h_1^{\alpha_1} \cdots h_n^{\alpha_n}, \quad |\alpha| = \alpha_1 + \cdots + \alpha_n.$$

Thus if we wanted to unfurl the definition we could instead write

$$f(x+h) = \sum_{\alpha_1, \ldots, \alpha_n \le k} \frac{1}{\alpha_1! \cdots \alpha_n!} \frac{\partial^{\alpha_1}}{\partial x^{\alpha_1}} \cdots \frac{\partial^{\alpha_n} f}{\partial x^{\alpha_n}}(x) \cdot h_1^{\alpha_1} \cdots h_n^{\alpha_n} + o(h).$$

Definition 4.2 — A function $f\colon U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is C^∞ or **smooth** or **infinitely differentiable** if its k-th derivative exists for all k. We say f is analytic if f is smooth and for all $x \in U$, there is an $r > 0$ such that $\|h\| < r$ implies that the Taylor polynomials around x within this ball converge to x.

Exercise 4.3. Show that the inverse/implicit function theorems produce C^k or C^∞ or analytic functions if they are given respectively C^k/C^∞/analytic functions. (woah analytic??)

4.2 Integration over rectangles!!

Fuck wait Riemann sums again nooooooooooo

Recall that a **partition** $\mathcal{P}$ of an interval $[a, b]$ is a finite collection of points $a = t_0 < t_1 < \cdots < t_k = b$. We would like to generalize this to arbitrary boxes so that we can talk about Riemann sums over arbitrary boxes.

Definition 4.4 — A **partition** $\mathcal{P}$ of the rectangle Q is an n-tiuple $(\mathcal{P}_1, \ldots, \mathcal{P}_n)$ where $\mathcal{P}_j$ is a partition of $[a_j, b_j]$.

explain intuition

Let $f\colon Q \to \mathbb{R}$ be a bounded function and define

$$m_Q(f) = \inf_{x\in Q} f(x), \quad M_Q(f) = \sup_{x\in Q} f(x).$$

Then, given a partition $\mathcal{P}$ of Q, the **lower** and **upper Riemann sums** are respectively

$$L(f, \mathcal{P}) = \sum_R m_R(f) \cdot |R|, \quad U(f, \mathcal{P}) = \sum_R M_R(f) \cdot |R|$$

where the sums are over all subrectangles R of the partition and $|R|$ is the volume of the box R.

Lemma 4.5. Let $\mathcal{P}'$ be a refinement of a partition $\mathcal{P}$ of a box Q. Then,

$$L(f, \mathcal{P}) \le L(f, \mathcal{P}'), \quad U(f, \mathcal{P}) \ge U(f, \mathcal{P}')$$

Proof. Look at monotonicity of infimum/supremum. □

Lemma 4.6. If $\mathcal{P}$ and $\mathcal{P}'$ are two partitions of Q then

$$L(f, \mathcal{P}) \le U(f, \mathcal{P}').$$

Proof. Look at the common refinement of the two partitions. □

Definition 4.7 (Lower/upper Riemann integrals) — The lower Riemann integral of a bounded $f\colon Q \to \mathbb{R}$ (Q box) is

$$\underline{\int_Q} f := \sup_{\mathcal{P}} L(f, \mathcal{P})$$

and the upper Riemann integral is

$$\overline{\int_Q} f := \inf_{\mathcal{P}} L(f, \mathcal{P})$$

The lower integral is always at most as large as the upper integral; if they are equal, then we just write

$$\int_Q f := \overline{\int}_Q f = \underline{\int}_Q f$$

and say that f is Riemann integrable.

Lemma 4.8. A bounded function $f\colon Q \to \mathbb{R}$ is Riemann integrable if and only if for every $\varepsilon > 0$ there is a partition $\mathcal{P}$ of Q such that

$$U(f,\mathcal{P}) - L(f,\mathcal{P}) < \varepsilon.$$

5 Week 5: 2/24 and 2/26

We are interested in describing what Riemann integrable functions look like.

Theorem 5.1. Let $Q \subseteq \mathbb{R}^n$ be a rectangle and $f\colon Q \to \mathbb{R}$ bounded. Then f is (Riemann) integrable if and only if the set $D \subseteq Q$ on which f is discontinuous has measure zero.

Given bounded $f\colon Q \to \mathbb{R}$ and $x \in Q$, let

$$A_{\delta,x} = \{f(y) : y \in Q, \|y - x\| < \delta\}.$$

The **oscillation** of f at $x \in Q$ is

$$\operatorname{osc}(f,x) = \inf_{\delta>0} \sup A_{\delta,x} - \inf A_{\delta,x}.$$

Lemma 5.2. The oscillation of f at a point x is zero if and only if f is continuous at x.

The proof comes from just unraveling the definitions.

Proof of Theorem 5.1. Suppose f is Riemann integrable over Q and let D be its set of discontinuities. We can then write

$$D_m = \left\{x \in Q : \operatorname{osc}(f,x) < \frac{1}{m}\right\}$$

so that $D = \bigcup_{m=1}^{\infty} D_m$, so it suffices to prove that the D_m each have measure zero.

Let $\varepsilon > 0$ and pick a partition $\mathcal{P}$ of Q such that $U(f,\mathcal{P}) - L(f,\mathcal{P}) < \varepsilon$. Additionally, let $D'_m \subseteq D_m$ be the subset of D_m contained in the boundary of some subrectangle R of the partition of Q; we can ignore those points because they already lie in a set of measure zero (because boundaries of rectangles have measure zero). Thus let $D''_m = D_m \setminus D'_m$.

Let $R_1, \ldots, R_N \subseteq Q$ be the rectangles of $\mathcal{P}$ that contain some points of D''_m. For each $x \in D''_m$, if i is the index such that $x \in R_i$, then there exists $\delta > 0$ for which

$$B_\delta(x) \subseteq \operatorname{Int} R_i.$$

Then,

$$\frac{1}{m} \leq \operatorname{osc}(f,x) \leq \sup A_{\delta,x} - \inf A_{\delta,x} \leq M_{R_i}(f) - m_{R_i}(f).$$

In particular,

$$\sum_{i=1}^{N} \frac{1}{m}|R_i| \leq \sum_{i=1}^{N} (M_{R_i}(f) - m_{R_i}(f))|R_i| \leq U(f,\mathcal{P}) - L(f,\mathcal{P}) < \varepsilon,$$

which is enough to imply that D''_m is a null set because the R_i together cover D''_m and their total volume is at most $m\varepsilon$, which can get arbitrarily small. $\square$